

Conditional Variational Autoencoder (CVAE)

Azizul Hoque CE23B004

The objective of this model is to implement a generative framework for seismic intensity measures (IMs). Unlike neural networks, the CVAE maps complex, high-dimensional seismic spectra into a low-dimensional "latent space". By conditioning this latent space on earthquake parameters (Magnitude, Distance, etc.), the model can generate a distribution of possible ground motions, capturing both the mean behavior and the inherent uncertainty of seismic events.

Dataset and Preprocessing

The pipeline processes the NGA-Subduction Mega-Flatfile, which initially contains 71,340 earthquake records. The model identifies 105 distinct spectral acceleration periods ranging from $T = 0.010$ s to $T = 10.0$ s.

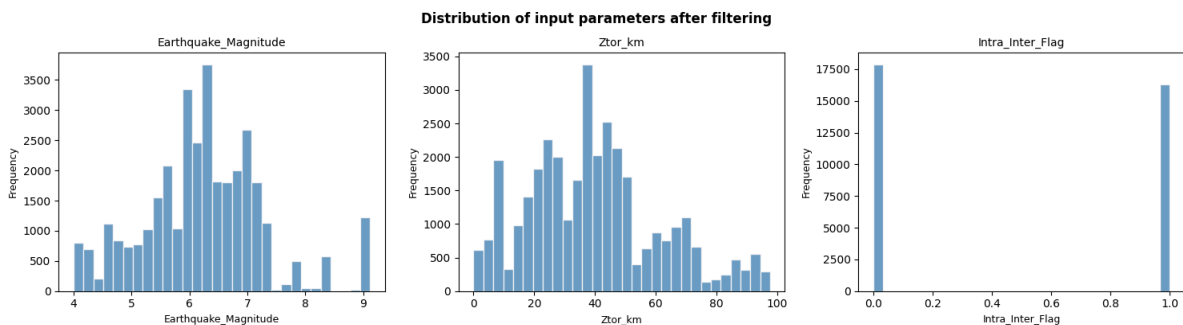
The data is categorized into primary inputs and target outputs:

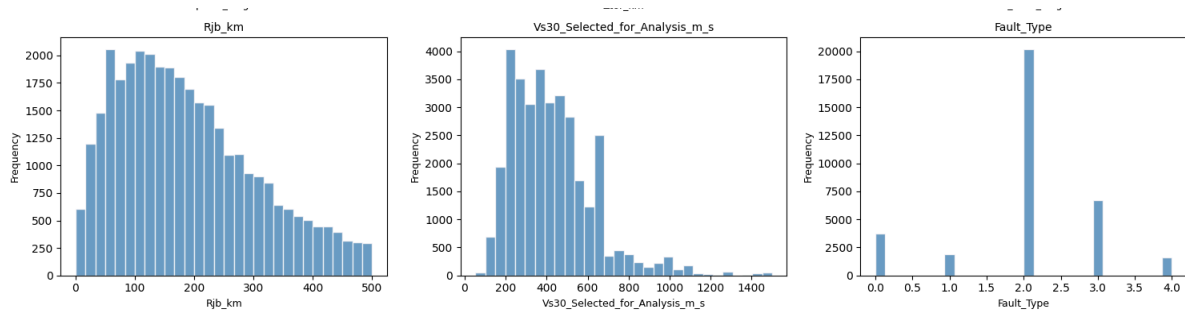
- **Primary Input Features:** These include Earthquake Magnitude M_w , Depth to top of rupture Z_{tor} , Intra/Inter-slab flag, Joyner-Boore distance R_{jb} , V_{s30} , and Fault Type.
- **Target Output Variables:** The pipeline targets Peak Ground Acceleration (PGA), Peak Ground Velocity (PGV), and the 105 identified Spectral Acceleration $[Sa(T)]$ periods.

Data Filtering:

- **Seismological Constraints:** Records are limited to magnitudes $M_w > 4$, rupture depths $0 < Z_{tor} < 100$ km, and distances $0 < R_{jb} < 500$ km.
- **Site and Event Classification:** Only records with Intra/Inter-slab flags of 0 or 1 are retained to exclude ambiguous events, and site conditions are limited to $V_{s30} < 1500$ m/s.
- **Intensity Measure Validation:** To prevent errors during log-transformation, the pipeline removes any records where PGA, PGV, or any of the 105 spectral acceleration values are non-positive or exceed 5g.

After applying these scalar and spectral filters, 37,117 records are removed, leaving a final clean dataset of **34,223 records**.





Feature Engineering

To better capture the non-linear physics of ground motion, five engineered features are added to the input vector:

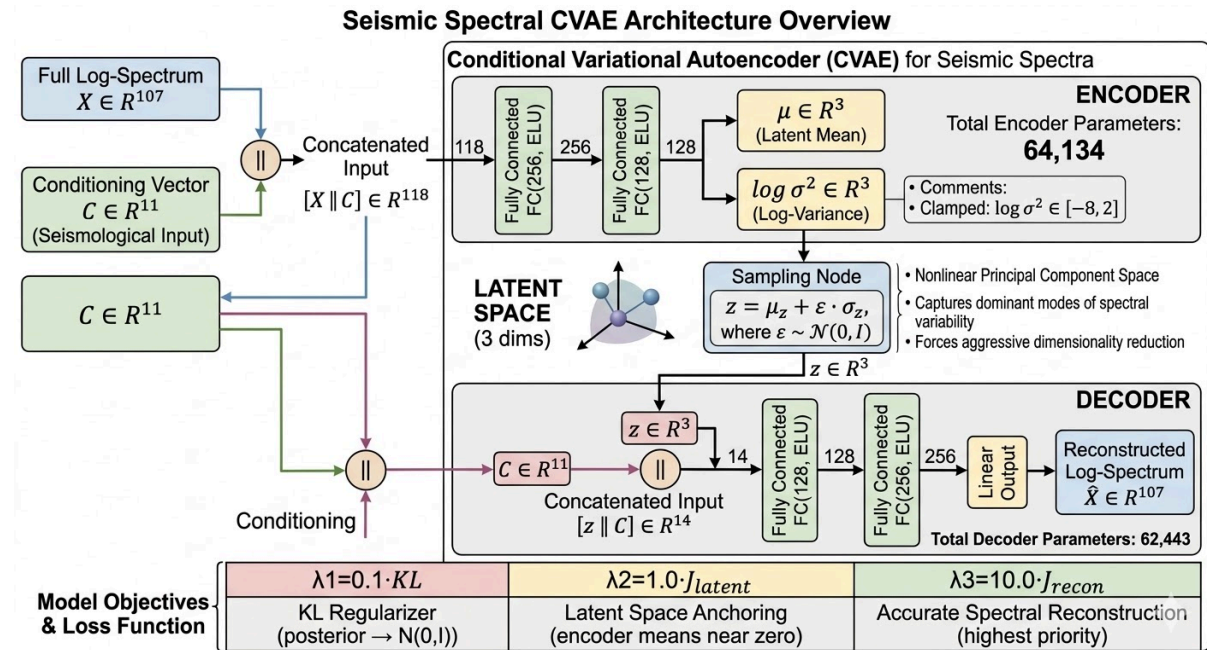
- **Magnitude Scaling:** A Magnitude_Squared (M_w^2) term is calculated to account for non-linear magnitude scaling.
- **Log-Transformations:** Natural log transforms are applied to distance (Log_Rjb), site conditions (Log_Vs30), and depth (Log_Ztor) with a +1 offset to handle surface ruptures).
- **Interactions:** A Magnitude-Distance interaction term (M_w times $\ln(R_{jb})$) is engineered to model the dependence of distance attenuation on earthquake size.

This results in a total of **11 refined input features**.

Feature	Description	Explanation
Earthquake Magnitude (M_w)	Moment magnitude of the earthquake	Bigger quakes = stronger shaking
Z_tor (km)	Depth to top of the rupture zone	Shallow ruptures cause stronger shaking at surface
Intra/Inter Flag	Type of subduction event: intraplate (0) or interface (1)	These two event types behave very differently seismically
R_jb (km)	Horizontal distance from site	Shaking decays with distance
Vs_30 (m/s)	Average shear-wave velocity of top 30 m of soil	Soft soils amplify shaking; rock does not
Fault Type	Type of fault mechanism	Affects radiation pattern of seismic waves
Magnitude Squared	M_w^2	Captures the nonlinear (curved) relationship
Log(R_{jb})	Natural log of distance	Distance attenuation follows a log-linear trend
Log(Z_{tor})	Log of rupture depth	Linearizes the depth scaling
Log(V_{s30})	Log of site velocity	Site amplification is approximately log-linear
$M_w * \text{Log}(R_{jb})$	Interaction term	Captures that large magnitude quakes attenuate differently over distance

Model Architecture

The CVAE consists of two symmetric networks, an encoder and a decoder, both conditioned on the seismological input vector $\mathbf{C}$ (11 features), with a 3-dimensional latent space representing the principal spectral components.



Encoder ($118 \rightarrow [256, 128] \rightarrow \mu, \sigma \in R^3$): receives the concatenated input $[X(107) \parallel C(11)] = 118$ dimensions and compresses it through two fully-connected ELU layers ($256 \rightarrow 128$ neurons) into two separate linear projections producing the latent mean μ and log-variance $\log \sigma^2$, each of dimension 3. The log-variance is clamped to $[-8, 2]$ for numerical stability. Total encoder parameters: **64,134**.

Latent space (3 dims): the 3-dimensional bottleneck acts as a nonlinear principal component space, capturing the dominant modes of spectral variability. A sample is drawn via the reparameterisation trick: $z = \mu_z + \epsilon \cdot \sigma_z$, $\epsilon \sim \mathcal{N}(1,0)$, keeping the operation differentiable. The small latent dimension (3 vs 107 outputs) enforces aggressive dimensionality reduction, forcing the model to learn only the most physically meaningful spectral features.

Decoder ($14 \rightarrow [128, 256] \rightarrow 107$): receives the concatenated $[z(3) \parallel C(11)] = 14$ dimensions and reconstructs the full 107-dimensional log-spectrum through two ELU layers ($128 \rightarrow 256$) followed by a linear output. The mirrored expanding architecture ($128 \rightarrow 256$) reverses the encoder's compression, progressively recovering spectral detail. Total decoder parameters: **62,443**.

Loss $\lambda_1 = 0.1 \cdot KL + \lambda_2 = 1.0 \cdot J_{latent} + \lambda_3 = 10.0 \cdot J_{recon}$: the three terms balance competing objectives — KL regularises the latent posterior toward $\mathcal{N}(0, I)$, J_{latent} anchors encoder means near zero for a structured space, and J_{recon} (weighted 10 $\times$) prioritises accurate spectral reconstruction above all else.

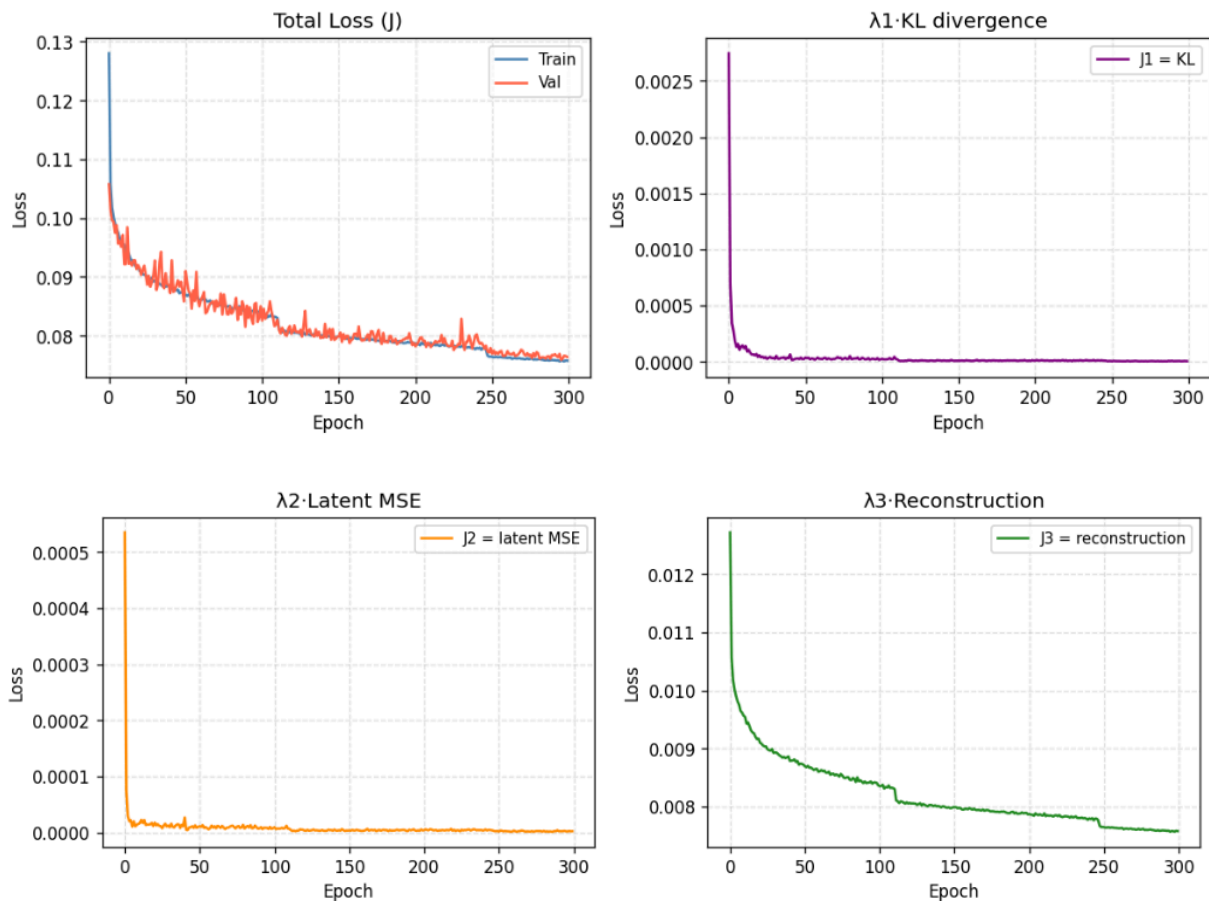
Model Training

Optimiser: Adam updates parameters with an initial learning rate of 0.001. Its adaptive per-parameter moments suit the varied gradient magnitudes from the three loss components. Gradient norms are clipped to 5.0 before each update to prevent early training instability caused by large KL term gradients.

Learning Rate Schedule: A ReduceLROnPlateau scheduler halves the learning rate if validation loss stalls for 20 epochs, with a minimum floor of 1×10^{-5} . Decays occurred near epoch 125 (0.001 to 5×10^{-4}) and epoch 225 (5×10^{-4} to 2.5×10^{-4}). Each step allows finer local minimum exploration, securing incremental gains.

Early Stopping: To prevent overfitting, training stops if validation loss does not improve for 40 consecutive epochs (patience = 40). The best encoder-decoder weights are saved and restored. This run completed all 300 epochs without triggering early stopping, suggesting ongoing learning.

Reparameterisation Trick: To enable gradient-based training through the stochastic latent sampling step, the latent code is not sampled directly. Instead, it is expressed as the posterior mean plus the posterior standard deviation multiplied by an independent noise vector drawn from a standard normal distribution. This reparameterisation decouples the randomness from the learnable parameters, making the gradient of the loss with respect to the encoder parameters well-defined and computable via standard backpropagation.



Inferences

Total Loss (Train vs Validation) Both losses drop steeply within the first 30 epochs and track closely throughout, with negligible train-validation gap at convergence (train ≈ 0.0758 , val ≈ 0.0764). This confirms stable learning with no overfitting. The two visible step-like plateaus near epochs 125 and 225 correspond directly to the learning rate decay events, after each of which loss resumes a modest downward trend.

J1 — KL Divergence Peaks sharply at epoch 1 and collapses to near-zero within the first 20–25 epochs, indicating the encoder quickly aligns its posterior with the standard normal prior. Its negligible value beyond epoch 25 means the KL term contributes no meaningful gradient signal for the majority of training.

J2 — Latent MSE Follows the same spike-and-collapse pattern as J1, confirming the posterior mean is tightly centred near the origin throughout training. Being the smallest of the three terms, it acts as a light supplementary regulariser rather than a dominant training signal.

J3 — Reconstruction Loss The dominant loss component across all 300 epochs, decreasing steadily from 0.0127 to 0.0076 without fully flattening, indicating the decoder continuously refines its predictions. Subtle slope changes at the LR decay epochs confirm the scheduler meaningfully aids reconstruction quality.

Extract PC's and Analyse

After training, the CVAE encoder maps each record into a 3-dimensional latent mean vector (μ), representing PC1, PC2, and PC3. The posterior mean μ is used as the ANN regression target rather than a sampled z , because it is the minimum-variance estimate of the latent code and avoids injecting stochastic noise into the targets. The KL divergence term encourages latent dimensions to be statistically independent, and the degree of actual uncorrelatedness is verified post-hoc. Correlation of PCs with seismological parameters checks whether the condition vector has fully absorbed the predictor information, low correlations confirm the PCs capture only residual aleatory variability unexplained by the known seismological predictors.

AFTER Encoding training set , Latent μ shape: (23956, 3) (3 PCs per record)

PC inter-correlation (should be ≈ 0):

	PC1	PC2	PC3
PC1	+1.0000	-0.6319	-0.6587
PC2	-0.6319	+1.0000	+0.0792
PC3	-0.6587	+0.0792	+1.0000

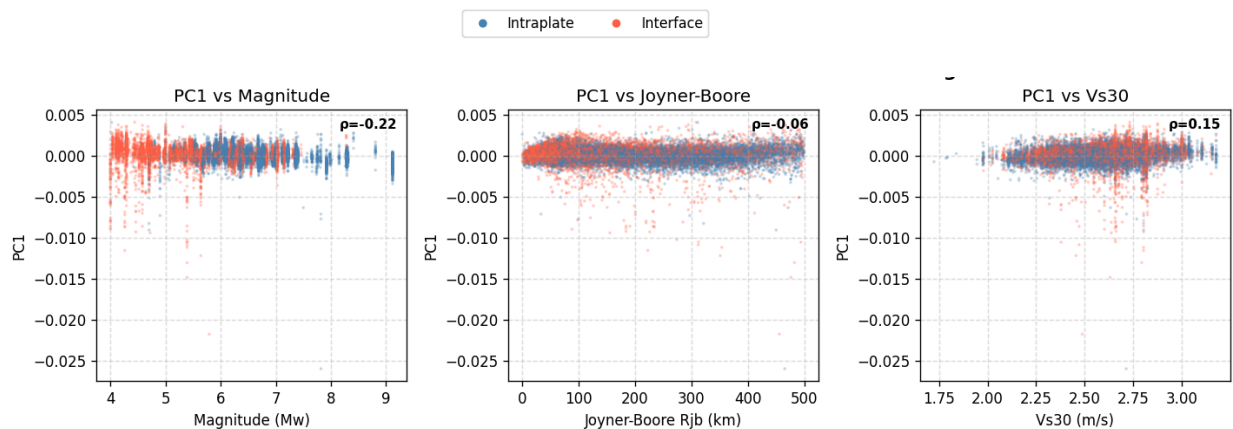
The variance explained by each Principal Component is as follows:

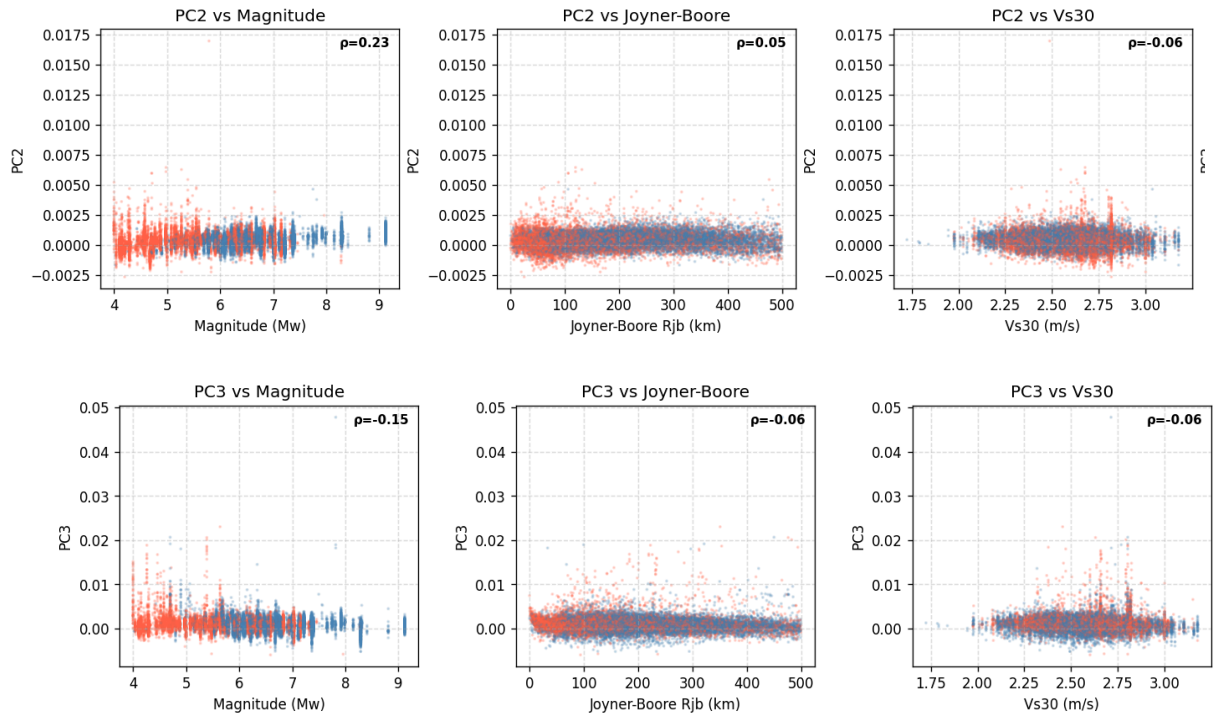
Principal Component	Variance Explained	Cumulative Variance
PC1	22.9%	22.9%
PC2	10.4%	33.3%
PC3	66.7%	100.0%

The correlation coefficients between the seismological parameters and the Principal Components are detailed below:

Parameter	PC1	PC2	PC3
Mw	-0.2157	+0.2255	-0.1458
Ztor	+0.2082	-0.2225	-0.0374
Flag	+0.0794	-0.0567	+0.0711
Rjb	-0.0560	+0.0513	-0.0568
Vs30	+0.1503	-0.0648	-0.0688
Fault	-0.0580	+0.0401	+0.0200
Mw^2	-0.2248	+0.2323	-0.1385
ln(R_jb)	-0.0600	+0.0643	-0.0656
ln(Z_tor+1)	+0.2140	-0.2685	-0.0495
ln(Vs_30)	+0.1491	-0.0645	-0.0628
Mw*ln(R_jb)	-0.1696	+0.1810	-0.1329

Latent Space Visualization





Inferences

PC1 Shows the strongest predictor dependencies among the three components. A mild negative correlation with magnitude ($\rho = -0.22$) and positive correlations with Vs30 ($\rho = +0.15$) and Ztor ($\rho = +0.21$) suggest PC1 partially captures site amplification and depth-related residual effects. Joyner-Boore distance shows negligible influence ($\rho = -0.06$), confirming attenuation is well handled by the condition vector.

PC2 Positively correlated with magnitude ($\rho = +0.23$) and negatively with Ztor ($\rho = -0.22$), indicating PC2 encodes complementary spectral shape variation related to source size and rupture depth. Distance and Vs30 show near-zero correlation, confirming these effects are absorbed by the conditioning.

PC3 Exhibits the weakest correlations across all four parameters (all below ± 0.15), consistent with it representing predominantly aleatory ground-motion variability rather than any systematic predictor trend. This supports PC3's role as the dominant variance-carrying component (66.7%) of pure within-event randomness.

Fault Type (Intraplate vs Interface) Blue (Intraplate) and red (Interface) points overlap substantially across all PC-parameter combinations with no visible systematic separation, suggesting the CVAE latent space does not strongly distinguish between fault mechanisms beyond what the condition vector already encodes.

Overall The PCs are largely free of strong seismological predictor dependence, particularly for distance and site parameters. The mild residual correlations with magnitude and depth are acceptable and consistent with known limitations of CVAE disentanglement under low KL weighting. The latent space is physically reasonable and suitable for ANN-based regression targeting.

ANN for Latent Prediction & Full Pipeline

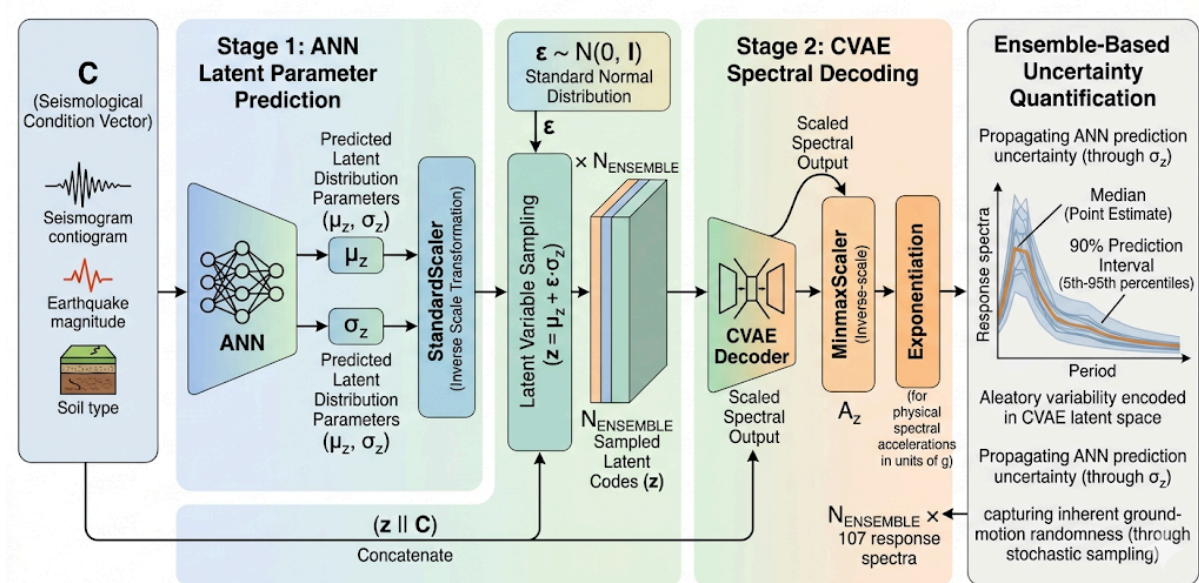
Model Architecture

The network uses a compact bottleneck: Input(11) $\rightarrow$ ELU(6) $\rightarrow$ ELU(3) $\rightarrow$ Output(6). The 11-dimensional input is compressed through two ELU-activated hidden layers before splitting into two output heads for latent mean (μ) and standard deviation (σ). ELU aids convergence by providing smooth, non-zero gradients for negative inputs. The sigma output uses Softplus + epsilon to guarantee positivity.

Target Normalisation: The concatenated latent targets [μ , σ] are standardised with a `StandardScaler` prior to training. This equalizes their contribution to the MSE loss, preventing any single principal component (PC) from dominating gradient updates.

Loss Function and Optimisation: The ANN minimizes the mean-squared error between predicted and CVAE-encoded latent parameters in the standardized space. The same Adam optimizer, `ReduceLROnPlateau` scheduler, gradient clipping, and early stopping strategy used for CVAE training are applied for consistent, stable optimization.

Full Prediction Pipeline



The predictive process is a **Two-Stage Generative Pipeline** where an Artificial Neural Network (ANN) and a Conditional Variational Autoencoder (CVAE) decoder are combined.

- ANN Prediction and Latent Parameter Transformation:** A raw seismological condition vector is first fed into the ANN, which predicts the latent distribution parameters (μ_z , σ_z). These parameters are then inverse-transformed using the stored `StandardScaler` to align them with the original CVAE latent scale.
- Ensemble Sampling and Decoding:** An ensemble of N latent codes is generated via the reparameterization trick: $z = \mu_z + \epsilon \cdot \sigma_z$, where ϵ is sampled from a standard normal distribution. Each sampled z , along with the condition vector, is then decoded by the CVAE decoder. This yields a scaled spectral output.

3. **Final Transformation:** The scaled output is inverse-transformed using the **MinMaxScaler** and subsequently exponentiated to recover the final physical spectral accelerations, expressed in units of g.

This procedure facilitates **Ensemble-Based Uncertainty Quantification**, as drawing multiple epsilon samples results in an ensemble of N_ENSEMBLE response spectra, rather than a single deterministic prediction.

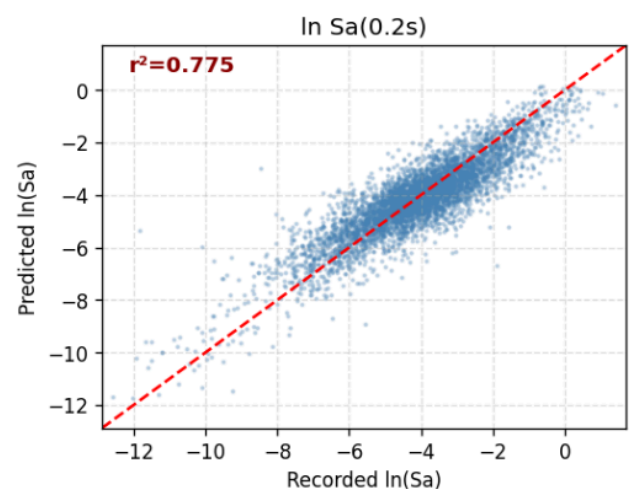
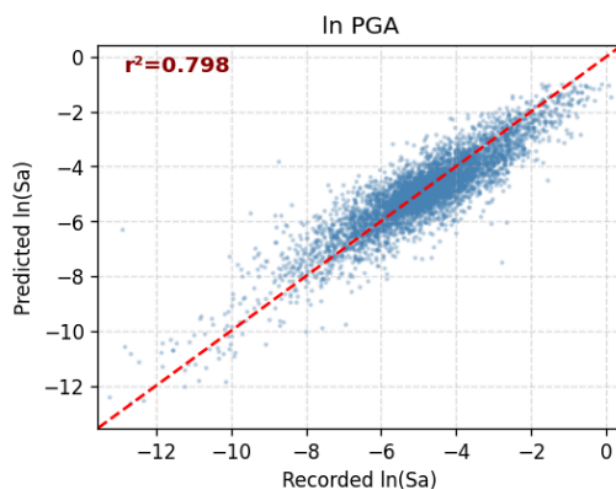
- **Point Estimate:** The median of the ensemble serves as the final point estimate.
- **Prediction Interval:** The 5th and 95th percentiles define the 90% prediction interval.

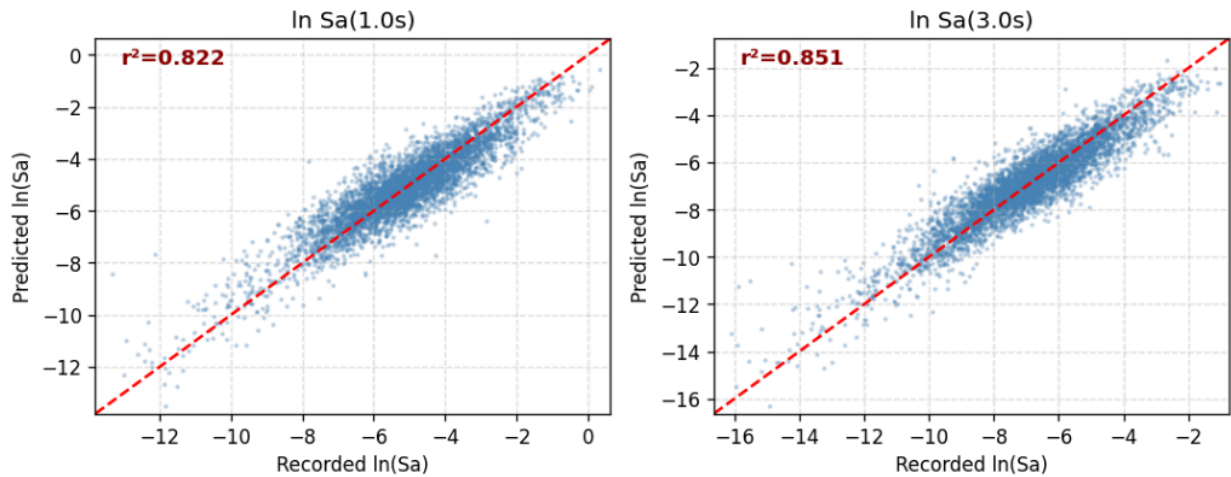
This ensemble approach effectively quantifies the **aleatory variability** inherent in the CVAE latent space by propagating both the uncertainty from the ANN prediction (captured by sigma_z and the intrinsic ground-motion randomness (via the stochastic sampling) into the final spectral predictions.

Observed vs Predicted Scatter Plots

Prediction of Test Set Results (Deterministic Mean)

Metric	PGA	Sa(0.2s)	Sa(1.0s)	Sa(3.0s)	Overall
Correlation Coefficient (R)	0.8931	0.8801	0.9065	0.9226	0.9484
Root Mean Square Error (RMSE)	0.7236	0.8033	0.7174	0.7547	0.7526
Mean Absolute Error (MAE)	0.5516	0.6163	0.5592	0.5863	0.5771
Mean Absolute Percentage Error (MAPE)	14.4364	28.3004	14.1362	9.4236	41.6378
Performance Period (PP)	0.3026	0.2758	0.3338	0.3772	0.4663





Inferences

Overall Predictive Performance The model achieves R^2 values of 0.798 (PGA), 0.775 (Sa 0.2s), 0.822 (Sa 1.0s), and 0.851 (Sa 3.0s), indicating good to strong predictive skill across the spectral range. Performance improves progressively from short to long periods, suggesting the CVAE latent space captures long-period ground-motion variability more effectively.

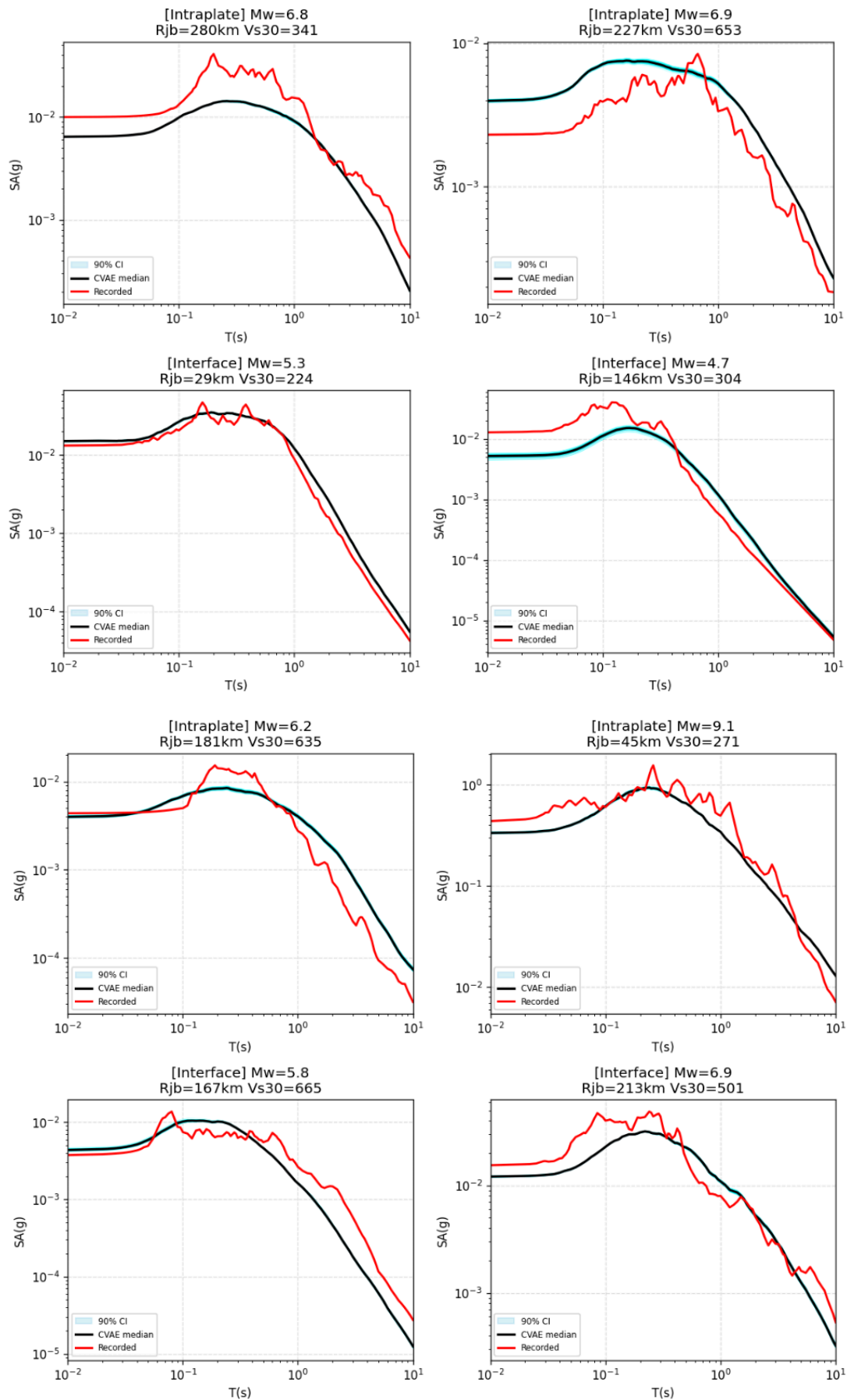
Scatter Around the 1:1 Line All four plots show predictions clustering reasonably along the ideal 1:1 dashed line with no significant systematic bias. The spread is broadly symmetric, confirming the model neither consistently over- nor under-predicts across the intensity range.

Larger Scatter at Short Periods Sa(0.2s) shows the lowest R^2 (0.775) and visibly wider scatter, reflecting the well-known difficulty in predicting short-period motions, which are more sensitive to local site effects and high-frequency source variability not fully captured by the 11-parameter condition vector.

Tail Behaviour At very low and very high $\ln(Sa)$ values, slight underprediction of extreme values is visible, which is common in regression-based ground-motion models due to data sparsity in the tails of the distribution.

Overall The CVAE+ANN pipeline produces physically consistent and well-calibrated median predictions across all spectral periods, with R^2 consistently above 0.77, validating the two-stage latent regression approach for subduction ground-motion modelling.

Ensemble Predictions- 100 Samples



Inferences

General Spectral Shape The CVAE median (black) reproduces the characteristic subduction response spectral shape across all eight scenarios, smooth broadband decay with period, broadly consistent with expected ground-motion behaviour for both intraplate and interface events.

90% Confidence Interval The cyan ensemble band is consistently narrow across all scenarios, indicating low aleatory variability in the model predictions. The tight CI reflects the compact latent space learned by the CVAE and confirms stable ensemble behaviour across 100 samples.

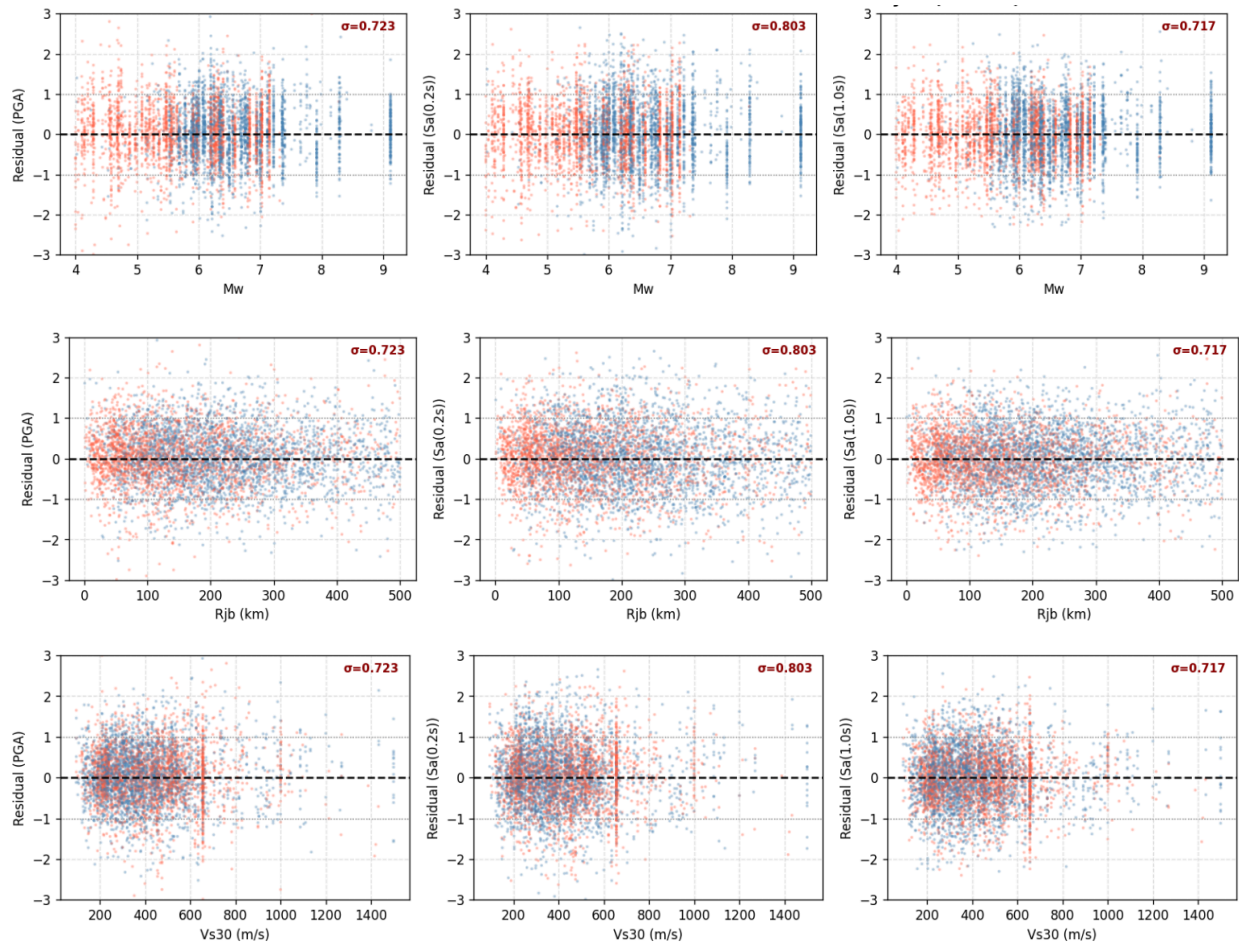
Recorded vs Predicted The recorded spectra (red) fall within or near the 90% CI in several cases, particularly for Interface events ($M_w=4.7$, $M_w=5.8$, $M_w=6.9$), confirming good model calibration at moderate magnitudes and distances. However, notable underpredictions are observed in short-period ranges for several Intraplate records, where the recorded spectra exceed the upper CI bound.

Intraplate Events The model systematically underpredicts high-frequency (short-period) amplitudes for intraplate scenarios, particularly at large distances ($R_{jb} = 280$ km, 227 km). This is expected, as intraplate records are fewer in the NGA-Subduction dataset, limiting the CVAE's ability to fully characterise their higher stress-drop characteristics.

Large Magnitude Event ($M_w=9.1$) For the large interface event, the median prediction tracks the recorded spectrum well across mid to long periods, with the recorded curve falling within the CI band, a strong validation of the model's performance for extreme magnitude scenarios.

Overall The CVAE+ANN pipeline produces physically plausible and well-shaped response spectra with reliable uncertainty bounds. Performance is strongest for interface events and long periods, while short-period intraplate predictions show residual underprediction, pointing to a known data imbalance limitation in the training set.

Residual Analysis



Inferences

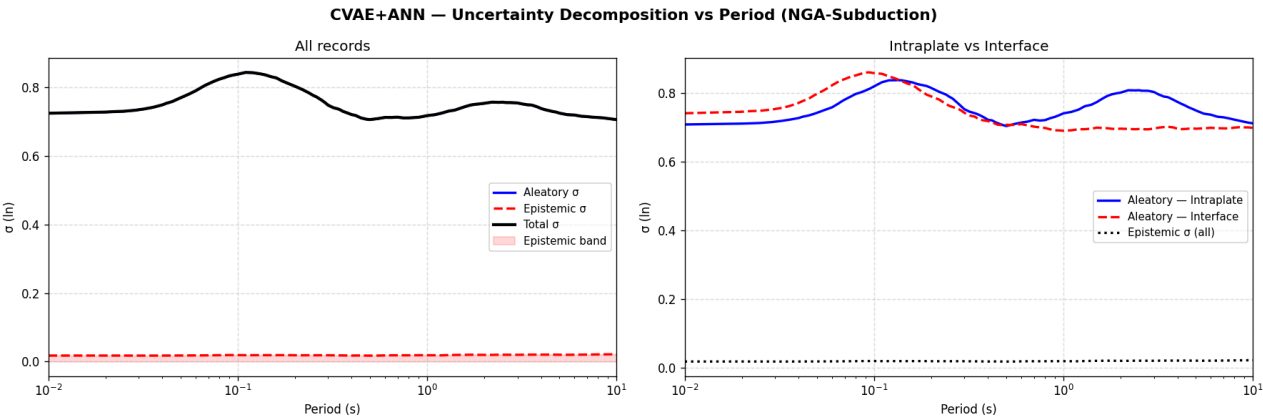
- The overall standard deviation (sigma) ranges from 0.717 (Sa 1.0s) to 0.803 (Sa 0.2s), typical for subduction zone models, with the lowest sigma at 1.0s confirming best performance at intermediate periods.
- Residuals are centred near zero across the full magnitude (Mw 4–9) and distance (0–500 km) ranges, indicating adequate capture of magnitude scaling, geometric spreading, and attenuation. Minor scatter increases at higher magnitudes and decreases at larger distances are consistent with natural data heteroscedasticity.
- No systematic trend is seen in residuals versus Vs30 (150–1500 m/s), confirming site amplification is reasonably captured. Higher scatter at low Vs30 reflects greater soft soil variability.
- Residuals for both intraplate and interface events are centred near zero; however, the marginally wider dispersion for intraplate residuals is consistent with their higher variability and sparser representation in the training data.
- The model is largely unbiased across all predictors, and the residual structure validates the CVAE+ANN pipeline as a well-calibrated ground-motion model for NGA-Subduction data, with variability primarily aleatory.

Epistemic Uncertainty Decomposition

Aleatory Uncertainty (σ_{aleat}): Represents inherent, irreducible randomness in ground motions. It is the standard deviation of residuals between observed spectral values and model predictions , calculated per period to reflect frequency variation.

Epistemic Uncertainty (σ_{epis}): Stems from the model's (ANN's) structural deficiency in mapping seismological conditions to the latent space. It is calculated using an **ensemble methodology**: N latent codes are sampled per record using the predicted mean and standard deviation. The variance across these measures the model's "lack of knowledge."

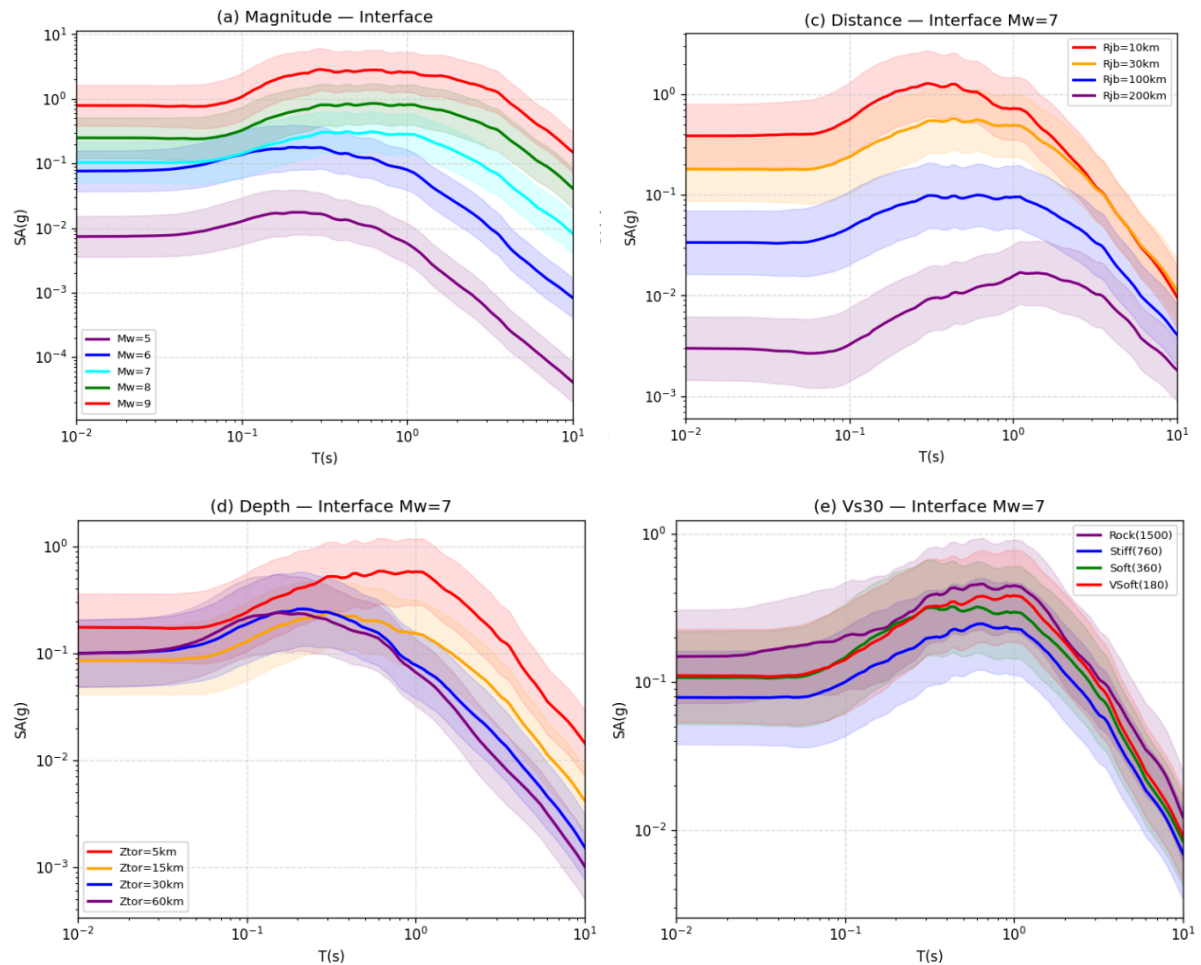
Uncertainty Type	Range of sigma (ln scale)
Aleatory	0.705 – 0.843
Epistemic	0.017 – 0.021
Total	0.706 – 0.843



Inferences

- **Dominant Aleatory Variability:** The total error Total sigma is primarily driven by irreducible, inherent ground motion randomness Aleatory sigma approx 0.7–0.85.
- **Minimal Epistemic Uncertainty:** The Epistemic sigma (model uncertainty) is very low and flat, indicating high model confidence. Further complexity or data of the same type is unlikely to significantly reduce the overall error.
- **Tectonic-Specific Trends:** Variability differs by earthquake type: Interface events show higher short-period (approx 0.1s) variability, while Intraplate events peak at long periods (approx 2–3s).
- **Frequency Sensitivity:** The CVAE finds high-frequency spectral accelerations (approx 0.1s) most difficult to predict, as shown by the peak in Total sigma.

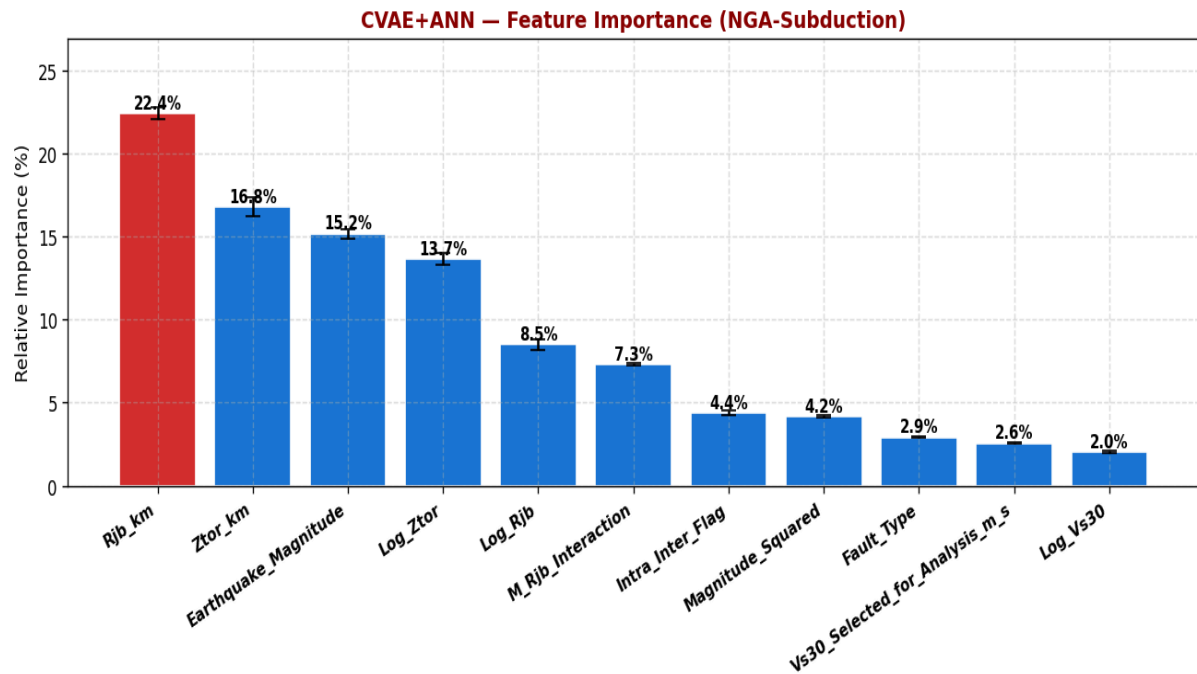
Parametric Study



Inferences

- **Magnitude (M_w):** As expected, higher magnitudes lead to higher Spectral Acceleration (SA) across all periods. The scaling is smooth and follows a clear hierarchy from $M_w = 5$ to $M_w = 9$.
- **Distance (R_{jb}):** Ground motion intensity decays significantly with increasing distance. The model accurately represents the geometric spreading and attenuation of seismic waves.
- **Depth (Z_{tor}):** Shallower earthquakes (< 5 km) produce much stronger surface shaking than deep events (> 60 km), particularly at longer periods where the energy is better preserved.
- **Site Effects (V_{s30}):** Softer soil sites (180 m/s) show higher amplification and a shift in the resonance peak compared to stiff rock sites (1500 m/s).
- **Tectonic Regime:** For the same magnitude, **Interface** events (red) generally produce higher spectral accelerations than **Intraplate** events (blue) across the entire frequency range.
- **Aleatory Variability:** The pm sigma shading is consistent across scenarios, confirming that the model captures the inherent "noise" and uncertainty of real-world ground motions while maintaining stable mean predictions.

Feature Importance



- **Distance is the Primary Driver:** **Rjb_km** (22.4%) is the most influential feature. This confirms that the geometric spreading of seismic waves is the single most important factor in determining ground motion intensity.
- **Depth and Magnitude are Crucial:** Source parameters like **Ztor_km** (16.8%) and **Earthquake_Magnitude** (15.2%) are the next most significant predictors. The model relies heavily on where the rupture started and how much energy it released.
- **Capturing Non-Linearity:** The inclusion of log-transformed variables (**Log_Ztor**, **Log_Rjb**) and interaction terms (**M_Rjb_Interaction**) among the top features shows the model is successfully modeling the non-linear physics of seismic attenuation.

Model Summary

Architecture

CVAE Encoder : (118)→[256, 128]→(μ,σ)∈R³

CVAE Decoder : (14)→[128, 256]→107

ANN : 11→[6, 3]→(μ_z,σ_z)∈R³

Loss weights : λ₁=0.1(KL) λ₂=1.0(latent) λ₃=10.0(recon)

Ensemble N : 100 samples per scenario

Performance

Test R² (overall) : 0.8994

Test RMSE : 0.75256

Test MAE : 0.57712

Aleatory σ (Sa) : 0.705–0.843 (ln)

Epistemic σ (Sa) : 0.017–0.021 (ln)

Total σ (Sa) : 0.706–0.843 (ln)